

VARDHAN KUMAR SIDDHANTHAM

LS6 1JY, LEEDS | 07774745934 | vardhan.ms.edu@gmail.com | <https://www.linkedin.com/in/svardhankumar/>

Technical Skills

- **Data Analysis & Visualization:** Python (Pandas, NumPy, Matplotlib, Seaborn, Plotly), R, SQL, Power BI, Tableau, Excel
- **Machine Learning & Deep Learning:** Classification, Clustering, Dimensionality Reduction, Ensemble Methods, Neural Networks (ANN, CNN, RNN), Autoencoders, GANs
- **NLP & Generative AI:** Text Analysis, NER, Transformers (BERT, GPT, LLaMA2, Mistral, Gemini), Agentic AI
- **Frameworks & Tools:** TensorFlow, PyTorch, Keras, Scikit-learn, XGBoost, LightGBM, SpaCy, NLTK, Gensim
- **Other Skills:** Model Evaluation, A/B Testing, Hyperparameter Tuning, AWS

Work Experience

Gen AI Engineer

09/2024 - 04/2025

Aadyam Solutions, Delhi - India

Technologies: Google Gemini LLM, RAG, FAISS, FastAPI, Python, Google Gemini Embeddings

- Built an AI-powered shopping assistant using Google Gemini LLM with RAG, enabling natural language queries and human-like responses for product discovery, comparisons, etc and achieving high response accuracy by updating the vector database with new product information and optimising embeddings for better contextual understanding.
- Implemented a high-speed vector search system using FAISS and Google Gemini embeddings, enabling efficient semantic search across a large product catalogue, reducing response time to ~100ms per query.
- Developed a backend with FastAPI and Python, integrating real-time data retrieval for pricing, availability, and returns.
- Automated customer support workflows, increasing engagement by 50% and reducing manual efforts by 80%.

Product Analyst

03/2022 - 09/2023

The Shadox Merchandise, Hyderabad - India

- Led Data-Driven Sentiment Analysis Project on Customer Reviews by collecting and analysing over 10,000 customer reviews sourced from an internal feedback platform using web scraping (Scrapy) and API integration for data acquisition, then cleaned and processed data using Pandas, NLTK, and SpaCy.
- Applied Vader for sentiment scoring and trained machine learning models (Logistic Regression, Random Forest) using Scikit-learn, attaining accuracy (85%), precision (82%), recall (80%), and F1-score (81%).
- Incorporated Tableau alongside Matplotlib and Seaborn to create interactive dashboards and visualizations, revealing sentiment trends and providing actionable insights to enhance customer performance.

Education

MSc Data Science and Analytics, University of Leeds, United Kingdom (Merit)

09/2023 - 09/2024

- **Subjects:** Data Science, Machine Learning, Deep Learning, Statistical Theory and Methods, Business Analytics and Decision Science, Learning skills through Case Studies, Statistical Learning.

Bachelor of Technology, Computer Science, K L University, India (Distinction)

07/2018 - 05/2022

Projects

MSc Dissertation – Predictive Modeling of Student Satisfaction (NSS): Engineered a multi-layer deep neural network using TensorFlow with ReLU activations and Adam optimizer, earning 85%+ accuracy on Q27 satisfaction prediction from 20,000+ NSS records; applied cross-regional generalization using data from Wales, Scotland, and NI.

ICO Success Prediction Model: Developed and tuned multiple classification models (SVM, Random Forest, GBM, Logistic Regression) on 2,767 ICO datasets; performed extensive feature engineering and hyperparameter optimization, achieving 75% accuracy and 0.74 ROC-AUC on the best-performing SVM.

Image Classification – CIFAR-100 Dataset: Designed and trained a deep CNN architecture with normalization, data augmentation (random cropping/flipping), learning rate scheduling, and dropout regularization, achieving 75%+ accuracy and improving validation performance by 10%.

Publication

Detection of Plant Diseases Using Convolutional Neural Networks: Developed a CNN in TensorFlow, achieving 94% accuracy on the PlantVillage dataset and proposed mobile deployment for real-time crop disease detection.